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The Global Workspace Theory of Consciousness

Predictions and Results

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Introduction

Scientific studies of consciousness depart from philosophical explorations in one major respect: They must treat consciousness as an observable variable. We therefore study the differences between attended versus unattended events, masked versus conscious visual stimuli, the waking state versus sleep, and many other plausible comparison cases. Brain damage shows surprising dissociations between similar conscious and unconscious functions in blindsight, neglect, and between the ventral and dorsal streams of cortical vision. Experimental techniques like binocular rivalry and the attentional blink give us yet another set of conscious/unconscious comparison cases.

It is clear that “consciousness as a variable” makes a huge difference in mental life. Being conscious of “these words” enables many cognitive processes that are not enabled by the unconscious presentation of the same stimuli using backward masking. Nevertheless, the unconscious word presentation does activate cortical word regions, at lower levels and with less oscillatory resonance.

The challenge of teasing out an observable variable from the tangle of incidental facts is common in the history of science. Variables like gravity, air pressure, and (in biology) the nature of a species, all show a similar history. If we only have one instance of something, we cannot study its causal conditions and effects. Intuitively, a concept like “air pressure” may seem like one “thing,” but with vacuum pumps and barometers, air pressure becomes a continuous variable, and we can study its network of relationships.

For the last two decades I have called this approach “contrastive analysis” (e.g., Baars 1983, 1988, 2015) but it is of course just the experimental method applied to the study of consciousness. Contrastive analysis is quite different from philosophical debates, which traditionally begin with introspective reports without unconscious comparisons. Contrastive analysis always involves commensurable inferences, thereby avoiding the problem of unconscious self-observation.

The history of science shows many similar problems, where some existing concept was shown to be a variable and therefore observable “as” a variable, so that its causes and effects could be studied. “Air pressure” at sea level is one famous example. Naively, we tend to perceive air pressure as a single entity. But when we use a mercury column and compare the air pressure at two heights we see reliable differences.

The crucial step is to realize that the “weight” of the air around us is not a constant, but a variable with a true zero point.

Again, naively we tend to believe that gravity is a constant. Newton proposed that other bodies in space must have gravitational fields proportional to their masses. The philosopher George Berkeley rightly asked how this invisible force could possibly keep the earth in orbit around the sun. Newton did not know the answer, but was able to infer degrees of gravity as a function of the mass of the attracting objects. The real nature of gravity was not confirmed until the twenty-first century, when the existence of gravity particles (gravitons) was verified. Gravity may seem “obvious” in the twenty-first century, but earlier it was truly mysterious.

Regarding consciousness, for many years the burden of proof was placed on scientists to show that consciousness could be studied empirically (e.g., Dietrich & Hardcastle 2005). But it was only late in the twentieth century that empirical psychologists and brain scientists became generally convinced of the very wide spread of unconscious “intelligent” cognitive processes. By the 1970s Helmholtz’ claim about unconscious inferences (*unbewusste Schlüssen*) in visual perception finally commanded a consensus, and a myriad other unconscious examples of information processing emerged. With high-resolution brain imaging, as in direct electrode recordings in cortex, we can now observe visuotopical maps in cortex, linked by neuronal spike timing and by population oscillations. Inferotemporal facial maps activate coherently when humans look at a face, either consciously or unconsciously. Methods like binocular rivalry allow us to see both conscious and unconscious cortical activation (Dehaene, Kerszberg, & Changeux 1998; Scheinberg & Logothetis 1997).

Historically, evolving scientific concepts like “heat” do not begin with answers but questions. Adequate definitions come much later. Thermodynamics defined our concept of physical heat around 1900, but incomplete measures of heat were studied by Galileo, Fahrenheit, and others. Science often begins with operational definitions, not conceptual ones.

Even behaviorists agreed on an operational definition of consciousness, namely “verbal report.” More broadly, “accurate voluntary report” is still the most widely used operational index for conscious percepts, inner speech, and recall. This evidence is plentiful, as shown by William James’ magnum opus, *The Principles of Psychology* (1890). Today we can see a steady rise in empirical articles that cite consciousness or closely related terms, currently more than 5,000 articles per year. That number has been climbing consistently for decades (2003).

A number of scholars raise questions about the causal role of consciousness. However, that debate is also dependent on one’s operational definitions. If one treats subjective experience metaphysically, as something fundamentally different from the physical world, a relationship between “mental” and “physical” events seems paradoxical. But if one treats consciousness naturalistically, as an empirical construct like any other, subjective experience simply becomes a brain representation that is supported in a certain way by certain brain regions, notably the visual components of the thalamocortical complex (Edelman & Tononi 2000; Baars, Ramsøy, & Laureys 2004). Conscious cognition is certainly private, but it is not a closed-off division of reality. There is no paradox in speaking of the hippocampus as a region that supports learning of conscious episodes, and there should be no difficulty in speaking of the thalamocortical complex as a brain structure that supports conscious perception (in contrast to the cerebellum, for example) (Edelman & Tononi 2000).

A number of philosophers have adopted such a naturalistic approach (Dennett 2001; Churchland 1989; Block 2009). Naturalists maintain, in effect, that there is no reason to doubt the empirical status of consciousness when it is carefully defined as an empirical construct, like memory or language, with a clear role in a nomological network of other natural constructs (Cronbach & Meehl 1955; Hempel 1965).

For a simple example, we can say a word mentally and let it fade; yet for about ten seconds after fading it can still be recalled. Our ability to retrieve the word accurately suggests that an unconscious trace must have been maintained for a little while. Both the conscious experience of the word and its unconscious memory involve brain events, many of which can be observed directly via brain recordings.

This method of contrastive analysis, comparing similar conscious and unconscious brain events, therefore does not raise the traditional mind-body puzzles. All comparisons are between publicly observable events and constructs inferred from them. Some of these events are reportable and some are not.

We therefore have the makings of a controlled experiment on the role of consciousness in immediate memory. We can ask: “What is the effect of being conscious of a word?” With the advent of brain imaging, it has turned out that conscious words show strikingly different brain activity compared to closely comparable unconscious words (e.g., Dehaene 2001; Baars 2002).

Conscious experience is a difference that *makes* a difference.

We might avoid the *word* “consciousness” in such experiments, and speak only about “reportable” vs. “unreportable” brain events. Behaviorists have argued for this approach (Stevens 1966). However, purely descriptive operationism loses the advantage of generalizability: if we applied it to memory, for example, we would have one kind of memory for button-presses, a different one for verbal reports, yet a third for confidence ratings, and as many more as we have behavioral measures to explain. We would end up with a swarm of measures. To avoid such unnecessary multiplication of concepts, the term “memory” refers to a coherent construct inferred over many observations, just as “gravity” does in physics or “species” in biology.

It is worth remembering that theoretical constructs in biology and physics have also been the subject of centuries of debate. Scientific consensus rarely comes in a single leap.

“Consciousness” refers to a construct that can be inferred from numerous brain and behavioral observations, ranging from verbal reports to waking EEG. We could call it by some arbitrary label *C*, but fortunately, over a wide range of conditions the concept also happens to correspond to our personal experiences. Researchers in perception and psychophysics have long known this, and have routinely studied themselves as subjects in their own experiments.

Under the proper conditions, objective and subjective sources of information converge with great precision. Indeed, we cannot have our eyes examined by an optometrist without making use of this fundamental discovery of early nineteenth century psychophysics.

While the term “consciousness” extends an everyday word to a precisely defined empirical construct, this is also historically common. Scientific terms like “force,” “matter,” “light,” “color,” and “sleep” began as everyday words that were refined over time.

Recently, new brain discoveries have fundamentally changed our understanding of “deep” or “slow-wave” sleep.

Historically, inductive science simply adapts to new discoveries as they become credible beyond reasonable doubt. Readers who are used to thinking of consciousness in purely philosophical terms may therefore need to adjust to its naturalistic use as it is emerging in neuropsychology. Here we will use it like any other psychobiological concept.

Note that contrastive analysis does not evade the question of phenomenology. On the contrary, experiential reports are involved in each contrastive pair of phenomena. Binocular rivalry and binocular integration were discovered in the eighteenth century or even earlier. Today, a large number of experiments on conscious vision use binocular rivalry as a basic technique. With brain recording methods we can now compare a reportable stream of visual input to a physically identical unreportable stream. Subjects themselves spontaneously use the word “conscious” and “unconscious” for those conditions, and so do scientists and philosophers.

We can therefore directly observe both kinds of input streams on the activity of neurons in the visual brain (e.g., Kreiman, Fried, & Koch 2002). This kind of experimental comparison has now been pursued for decades, using a remarkable range of visual stimuli and even different biological species (for reviews, see Baars 2002; Baars et al. 2004; Frackowiack et al. 2004).

Rivalry experiments work well in species with overlapping visual fields, as in humans, macaques, and cats. Front-facing eyes using binocular integration and rivalry occur commonly in predators and in tree-dwelling species, and less commonly in herbivores like rabbits and horses, which have lateralized eyes, so that visual events to the side and even behind the animal’s body can be spotted. The probability of chasing vs. being chased seems to be the important difference in this respect. Rabbits gain more evolutionary benefits from detecting danger behind and to the side of the animal’s head, while cats are highly successful predators who benefit from excellent binocular depth perception and interocular integration.

Very similar integration and rivalry phenomena are known to exist in the other senses, such as hearing, the skin senses, active touch, olfaction, and vibrissal perception in rodents (a tactile system using nose hairs).

In species with subjective consciousness, like humans, such integration and competition phenomena give us useful experimental tools. Any theory that can account for this evidence tells us something about our visual consciousness. If we ever discover a coherent explanation for all the contrastive evidence, we will have a more complete theory (see Baars et al. 2013).

The Central Puzzle: Conscious Limits vs. Unconscious Vastness

Conscious cognition is closely associated with “limited capacity” aspects of the working brain. Limited capacity phenomena include immediate memory, the high selectivity of focal attention, coherent binding of perceptual features in consciousness, the limits of voluntary vs. automatic processing, and the fact that we cannot do two demanding actions at the same time.

If we look only at such “limited capacity” phenomena, the brain seems to be slow, serial, and error-prone. But observed directly the brain looks dramatically different.

The central nervous system is a massive beehive of neural assemblies, cells, layers, and connections, each flexibly specialized in some set of tasks, like analyzing visual shapes, maintaining body balance, or analyzing the syntax of this sentence. Most of these functions work in parallel and interactively, like one great “society of mind.” Together, the processing capacity of the unconscious brain society has not been estimated in a convincing way. The sheer number of neurons and synaptic connections, the signal carrying capacity of major pathways, and even the memory capacity of organs like the hippocampus are poorly understood. We do know that those unconscious “neural agents” must be very large in number, and that we have very little conscious access to them. On the positive side, brain studies have clarified many important aspects of unconscious functions. Specific neuronal models of unconscious “apps” are still tentative and incomplete. One of the most basic puzzles therefore, is: Why is the conscious aspect of the brain so radically limited when the unconscious part is so vast?

Brain Evidence for Vast Capacity of Unconscious Processes

Looking directly at the brain we see great orderly forests of neurons. A structure like the cortex has an estimated 80 billion cells. The interconnectedness of cortical neurons is remarkable; in less than seven steps we can reach from any neuron to any other. Cortical neurons combine into vast elegant fiber bundles running between thalamus and cortex; an estimated 600 million fibers connect the two hemispheres, and comparable numbers are bundled in great loops to connect distant points within each hemisphere. It is quite a beautiful and regular arrangement.

When we look at the brain directly we see a massively parallel-interactive organ: many processes and interactions happen at the same time. Most of them are not conscious.

The brain shows a *distributed* or *decentralized* style of functioning, in which the detailed work is done by billions of specialized neurons and neuronal groups, with no single command center.

There are interesting parallels with the cellular physiology of the body. Living organisms show decentralized control at the cellular level, with each cell being guided by its own genetic and epigenetic coding molecules, interacting with incoming and outgoing signals. While the major hormonal systems control vast numbers of cells, there is no single central executive for all body cells.

In spite of the brain’s distributed processing style, we can acquire voluntary (prefrontal) control of almost any part of the brain using conscious neurofeedback stimuli. To control alpha waves in cortex we merely sound a tone when alpha is detected in the EEG, and shortly the subject will be able to increase alpha at will, to some homeostatic limit. To control a single spinal motor neuron we can play back its electrical spikes over headphones; in a half-hour of practice subjects have been able to play drumrolls on single motor units. Neurons in the human temporal lobe can be trained to fire at will, given conscious perceptual feedback when the neuron fires (Cerf et al. 2014).

Even with subtle neurofeedback control we are not conscious of the details; rather, conscious feedback seems to recruit unconscious neuronal coalitions that handle the details. Conscious feedback control can be acquired over single neurons firing spikes, and even over larger populations rhythms.

Psychological Evidence for Vast Unconscious Capacity

We come to similar conclusions when we look at the mental lexicon of natural language, or at semantics or syntax. In a fraction of a second after glancing at a word in this chapter, your visual input is converted to a semantic code able to interpret its meaning. Going from words to meanings requires a large, mostly unconscious semantic lexicon. Educated speakers of English know about 100,000 words, generally with multiple word meanings. While we do not use all known words in everyday speech, we can understand each one very rapidly in context, almost as soon as it is presented.

But words are complicated things. The Oxford English Dictionary, for example, devotes 75,000 words to the many meanings of the word “set.” And yet, the abundant ambiguity of “set” is never experienced consciously. We can access one (or at most two) meanings at any given time.

Autobiographical Memory

The size of long-term memory is unknown, but we know that simply by paying attention to as many as 10,000 distinct pictures over several days, we can learn to recognize each of them without any attempt at memorizing. Stephen Kosslyn writes that “[t]he capacity of our visual memories is truly staggering; it is so large that it has yet to be estimated. ... Perhaps the most staggering results are reported by Standing (1973) who showed some of his subjects 10,000 arbitrarily selected pictures for 5 seconds each. ... His findings showed that there is no apparent upper bound on human memory for pictures. Moreover, with immediate recall, Standing estimated that if one million vivid pictures were shown to subjects, 986,300 would be recognized if one were tested immediately afterward; even after a delay, he estimates that 731,400 would be recalled” (Kosslyn 1980, p. 129).

Remarkable results like this are common when we simply ask people to choose between known and new pictures. Such recognition tests seem to work so well, because they re-present the original conscious experience in part or whole. Here the brain does a tremendous job of memory search, with little conscious effort.

We can get an everyday sense of this remarkable memory performance from the common experience of recognizing a film seen only once, many years ago, with a sudden sense of familiarity. Often we can even predict the next scene. It seems that the brain creates memories of the stream of experience merely by paying attention to it; but humans are always paying attention to new and interesting things, suggesting that our spontaneous autobiographical memory must be very large indeed. Once again we have a vast unconscious domain, and we gain access to it using conscious thought. Mere consciousness of an event helps to store it in memory, and when we experience it again we can voluntarily recognize one distinctive episode from many millions in memory.

In sum, it seems that the very limited stream of consciousness gives us access to billions of neurons, to a very large mental lexicon, and to an inestimably large source of autobiographical memories. Mere unaided consciousness may be sufficient to create rapid learning and accurate recognition. Consciousness is also needed to trigger a great number of automatic routines that combine to make up specific actions. All these striking effects of consciousness are unconscious. Conscious cognition may therefore be the gateway to the brain’s sources of unconscious knowledge and control.

Global Access: An Answer to the Puzzle of Limited Capacity?

The ancient theater metaphor of consciousness gives us one plausible way to think about this evidence. A theater combines limited events taking place on stage with a vast audience, just as consciousness involves very limited information that with access to a vast number of unconscious sources of knowledge.

Theater metaphors are helpful, but in the age of massively distributed connectivity and computation, as in the world-wide web, we have a more direct metaphor for the brain. Network mathematics for describing the web now apply directly to the “cortical connectome.”

Consciousness seems to be the publicity organ of the brain, or in Daniel Dennett’s phrase, it may be called “fame in the brain.” It is a capacity for accessing, disseminating, and exchanging information, and for exercising global coordination and control.

If we then allow for feedback signaling between cortical sources and sinks, “broadcasters” and “receivers,” we can model less-than-global coordination and control as well. The neuronal receivers can selectively ask for certain sources, or even suppress unwanted sources.

In the cortex, bidirectional signaling is the norm, so that neurons that receive signals in the cortico-thalamic complex also resonate adaptively to their sources. This could lead to potentially difficult nonlinearities, but direct studies of cortical signaling sometimes show predictable trajectories.

The idea that consciousness has an integrative function has a long history. Global Workspace Theory (GWT) suggests that the brain has a fleeting integrative capacity that enables access between functions that are otherwise separate. This makes sense in a brain that is viewed as a massive parallel set of highly specialized neuronal processors. In such a system coordination and control may take place by way of such a central information exchange, allowing some specialized processors – such as sensory regions in cortex – to distribute information to the system as a whole. This solution also works in large-scale computer architectures, which show typical “limited capacity” behavior when information flows by way of a global workspace. A sizable body of evidence suggests that consciousness is the primary agent of such a global access function in humans and other mammals (Baars 1983, 1988, 1997, 2002). Figure 16.1 shows a sketch of global access.

Table 16.1 shows seven general hypotheses suggested by GWT (Baars 1988, 1997, 2002; Baars & Franklin 2003; Baars et al. 2004). A number of scientists and philosophers now support some version of the global access hypothesis. This theoretical convergence is striking (e.g., Dennett 2001; Dehaene et al. 2001, 2003; Edelman & Tononi 2000; Freeman 2003; Cooney & Gazzaniga 2003). While theorists do not necessarily accept all features of GWT, they tend to agree on the global access hypothesis.

Global Workspace Theory and Specific GW Models

The most detailed exposition of GWT developed seven increasingly complex models (Baars 1988), all of which involve competition between global workspace inputs and widespread distribution to a large collection of unconscious specialized processors. Because more than a dozen different models have been developed under the umbrella of Global Workspace Theory, it is useful to make a clear distinction between the general

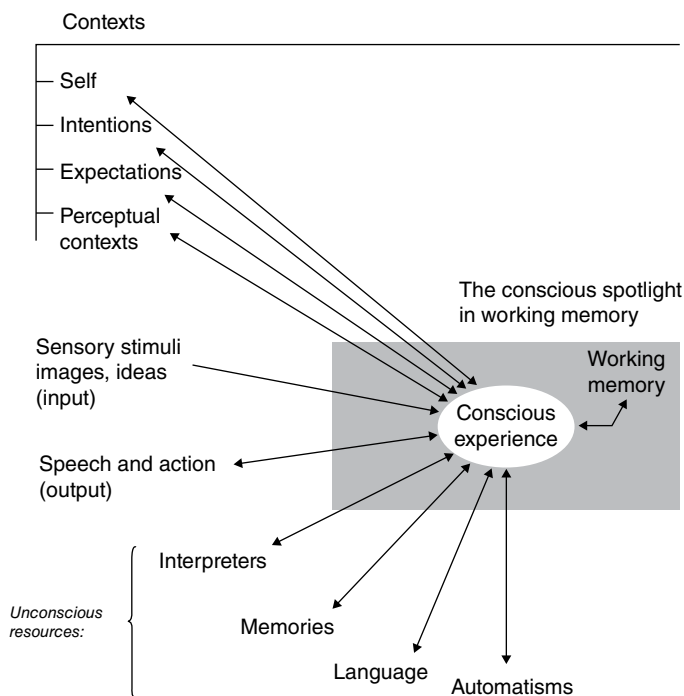


Figure 16.1 A schematic diagram of global access.

Table 16.1 Theoretical predictions: brain capacities enabled by conscious events. (Baars 1988, 2002)

- 1) Conscious perception enables access to widespread unconscious resources (“brain apps”) that are task-specific, distributed, and swarm-like.
- 2) Conscious perception, inner speech, and visual imagery enable working memory functions, including voluntary executive control.
- 3) Conscious events enable many kinds of learning: episodic and explicit learning, but also implicit and skill learning, which are not conscious as such, but are “consciously mediated”
- 4) Conscious neurofeedback enables voluntary control over motor functions, and indeed over many neuronal population and even single neurons (Cerf et al. 2013).
- 5) Conscious contents can evoke selective attention and be evoked by it.
- 6) Consciousness enables access to “self” – executive interpreters, located in prefrontal and medial parietal cortex.

features of GWT and specific GW Models (GWMs) that apply to experimental data sets, or which make use of specific theoretical mechanisms, such as neural net models or hybrid artificial intelligence architectures.

Table 16.2 shows specific empirical claims made by the seven cumulative GW Models of GWT, laid out in depth in *A Cognitive Theory of Consciousness* (Baars 1988). All but one of the major claims, which were highly controversial in 1988, are now supported by substantial bodies of neurocognitive evidence (see, e.g., Frackowiak et al. 2004; Baars et al. 2013).

Table 16.3 shows some Global Workspace Models developed by other authors. These are often neural net models tested by standard paradigms like visual backward masking and the attentional blink (studies by Dehaene and co-authors), as well as Franklin’s large-scale implementation of GW Theory in a large-scale computational model of

Table 16.2 Seven GW Models: predictions and results. (Baars 1988)

Model	Predictions	Evidence
GW Model 1. (p. 73)	Conscious percepts are always internally consistent, even if they derive from competing input streams. Consistent conscious end-products evoke global distribution and adaptation.	Global distribution: Many functional imaging studies (See Baars 2003). Internal consistency in visual object recognition appears to be established in temporal and parietal cortex.
GW Model 2. (p. 137)	Unconscious contexts shape conscious experiences.	Discovery of unconscious visuospatial maps in parietal cortex, which contextually influence the conscious visual content stream.
GW Model 3. (p. 177)	Conscious experience is informative – it always demands a degree of adaptation.	Dramatic loss of cortical activation with input redundancy – as in habituation and automaticity.
GW Model 4. (p. 225)	Conscious contents can establish goal contexts and facilitate spontaneous problem solving.	E.g., evoking unconscious social goals and knowledge by conscious cues.
GW Model 5. (p. 259)	Voluntary control is exercised by global distribution of conscious ideomotor images.	Frackowiak et al. summarize the relationship of prefrontal voluntary control widely activated by conscious, but not unconscious stimuli.
GW Model 6. (p. 301)	“Selective attention” involves voluntary control of access to consciousness.	Now also proposed by a number of other researchers.
GW Model 7. (p. 325)	“Self” as the dominant but unconscious context of experience and action.	Baars et al. 2004, frontoparietal hypometabolism in four types of unconscious states. Also overlap of executive regions with cortical areas activated by conscious, but not unconscious events. (Frackowiak et al. 2004)

Table 16.3 Dehaene’s Global Workspace Models: predictions and results.

Model	Predictions	Results
Dehaene & Changeux 2000. Reward-dependent learning in neuronal networks for planning and decision making.	Dorsolateral prefrontal and anterior cingulate cortex will be mobilized during acquisition of a novel task, effortful execution, and subsequent errors.	Higher BOLD activation in novel and effortful tasks.
Dehaene et al. 2001. Cerebral mechanisms of word masking and unconscious repetition priming.	Unconscious visual words will evoke local but not global cortical activity.	Backward-masked words activated left extrastriate, fusiform and precentral areas, while conscious words activated parietal, prefrontal regions as well.
Dehaene et al. 2003, A neuronal network model linking subjective reports and objective physiological data during conscious perception.	Global neuronal network model predicts sustained activity in primary visual area (V1), amplification of perceptual processing, correlation across distant regions, joint parietal, frontal, and cingulate activation, gamma-band oscillations, and P300 waveform.	All-or-none fMRI activation of frontoparietal regions in attentional blink paradigm, coinciding with subjective report.

cognition, called LIDA (Franklin 2000). In addition, Shanahan has developed GWT arguments for addressing widely debated cognitive science questions like modularity and the frame problem (Shanahan & Baars 2005). Baars & Franklin (2003) have also shown how the LIDA model of GWT can account for the role of conscious elements in cognitive Working Memory (Baddeley 2000).

Gaps and Limits in Current Work

The single biggest “gap in the literature” is between cognitive and brain evidence, but that gap is being bridged with extraordinary rapidity. Functional brain recordings of all kinds, ranging from single-cell studies in epileptic patients to functional MRI, have all contributed significant new insights. It no longer makes sense to separate brain and cognitive evidence or theory. GWT therefore must become a biocognitive theory (called Dynamic GWT, Baars et al. 2013). Clearly, much more work remains to be done along those lines.

Some fundamental conceptual questions need to be resolved. For example, the term “broadcasting” implies centrifugal flow of information from one center to many receivers. That is anatomically incorrect for the thalamocortical complex, where the great majority of links are bidirectional. This issue is discussed by Baars et al. (2013), by Freeman and Kozma, and by Edelman and Tononi. The advent of the world-wide web, which commonly employs two-way communication, gives us a closer analogy.

Critiques of Globalist Approaches

Some critics have wondered whether GWT is a “Cartesian Theater,” a *reductio ad absurdum* proposed by Daniel Dennett. If it is, that would refute GWT. But Dennett himself has never made such a criticism, and has indeed proposed his own slogan for GWT: “fame in the brain.” Those who wonder whether GWT is equivalent to a Cartesian Theater should therefore read Dennett (2001). The Cartesian *reductio* involves a point center “where everything comes together,” named after Descartes, who proposed that such a point center exists in the pineal gland. But of course there are no such point centers in Global Workspace Models as they were first developed by artificial intelligence researchers (Newell 1994; Franklin 2003). They are “virtual spaces,” implemented in a variety of formats. In real theaters many currents of information flow together – including the activities of playwrights, actors, directors, spot-light operators, stage hands, and of course, the attending audience.

The brain also has many areas of massive convergence and divergence, such as the reticular formation, thalamus, entorhinal cortex, planum temporale and indeed the entire thalamocortical system. None of these nexes involve point centers, so that they are not Cartesian absurdities. The critique of GWT as a Cartesian Theater is therefore mistaken.

The same point may be made for other standard philosophical critiques of scientific studies of consciousness. Most scientists and many philosophers reject such “in principle” arguments against empirical work. As we have already noted, GWT is based on comparisons between brain events that support conscious experiences vs.

those that do not, such as cortical vs. cerebellar activity. All of GWT is based on comparisons between clearly physical phenomena, though the conscious ones obviously are marked by phenomenological privacy from the perspective of the individual. Thus the standard critiques about qualia, the “Hard Problem,” and so on, simply do not apply.

Given the relative novelty of serious scientific work on this topic, there is a surprising amount of agreement among theorists so far. Most take a system-wide or architectural approach, since conscious experience is so clearly involved in all the major functions of the brain – perception, explicit cognition, learning and memory, imagery, voluntary control, problem-solving, emotional events and the like. A few claim that particular conscious contents involve only local cortical regions. The best known is Semir Zeki’s proposal for “microconscious” activity in small numbers of cortical neurons. Thus conscious color perception is proposed to be exclusively dependent on cortical area V4. This hypothesis depends on the finding of differential rise times for different visual regions in some studies (Zeki 2003).

It is not a new observation that different features of a stimulus become available at slightly different exposure times. It was a well-known observation in the nineteenth century. The question is whether such results imply a multitude of different types of consciousness, perhaps one for each kind of cortical feature cell. That would lead to thousands of micro-conscious brain regions.

From a GWT perspective such a very strong localizationist view involves a misunderstanding. Any brain interpretation of GWT must necessarily involve numerous specialized regions with slightly different rise times, because of conduction delays, reentrant loop thresholds, a changing balance of excitation and inhibition, attentional modulation, and numerous other variables. In addition, any interpretation of global distribution must involve thousands of populations of specialized cells, which can obviously be studied locally as well. Finding different operating characteristics of local regions in relation to consciousness is therefore no surprise. It certainly does not falsify GWT.

Indeed, a brain interpretation of GWT requires precisely the kind of convergent visual processing that we know to be happening in the flow from V1 to object representation regions in inferotemporal (IT) cortex. A landmark study by Sheinberg and Logothetis (1997) strongly suggests that for conscious object recognition, integration takes place in human area IT. From a GWT perspective, therefore, divergent broadcasting of visual contents might occur from IT to regions like hippocampus and Area 46, a visuotopically mapped area of prefrontal cortex. This could occur via cortico-cortical or cortico-thalamo-cortical connections (e.g., Dehaene 2001). Alternatively, widespread distribution could occur via horizontal spreading in Layer I of cortex (Freeman 2003). Some recent studies support the hypothesis that conscious contents involve field potentials in the alpha-to-gamma range. The predominance of alpha-to-gamma activity in conscious states may reflect resonant interaction between different brain regions. Such resonance may reflect both “binding” of different object features in the case of vision, and widespread distribution and recruitment of neuronal populations beyond visual cortex. These different mechanisms for “broadcasting” are not mutually exclusive. There may well be multiple brain mechanisms for GW-type functions. As Edelman and Galley (2001) maintain, such functional redundancy is a very general property of biological systems.

Summary and Conclusions

Global Workspace Theory suggests that conscious cognition involves numerous brain networks that cooperate and compete in solving problems. Consciousness is the gateway to the brain, enabling control even of single neurons and whole neuronal populations. None of these control functions becomes directly conscious, of course, but conscious feedback seems required to recruit local control by neuronal assemblies. In the metaphor of the theater, it seems as if each specialized audience member can decide locally whether or not to look at the bright spot on stage. Executive functions – like the director behind the scenes – are also largely unconscious, often using the actor in the spotlight of consciousness to recruit specific functions.

GW theory does not claim to be exhaustive. Like any theory, it requires a mass of evidence. No doubt our ignorance exceeds what we seem to understand by orders of magnitude. Nevertheless, the convergence of evidence and theory is encouraging so far.

See also 3 The hard problem of consciousness; 44 Integrated Information Theory of consciousness: some ontological considerations; 22 Daniel Dennett on the nature of consciousness; 31 Functionalism and qualia; 39 Conscious and unconscious perception; 40 Conscious and unconscious memory; 41 Consciousness of action.

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